DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-II

Design and Analysis of Algorithms

Class: B. E (CSE)
Semester: IV
Academic Year: 2025-2026

Max Marks: 30
Duration: 1 hr.30min
Date: 06-05-2026

		Marks	BT	CO
	PART A Answer All Questions (Q5 x 2 = 10 M)			
1. a)	Define Residual Network with an example.	2	2	3
b)	What is an optimal binary search tree?	2	2	3
c)	State Cook's theorem.	2	1	2
d)	Define the terms NP-Complete and NP-Hard.	2	2	2
e)	State the need of explicit and implicit constraints in Backtracking algorithm?	2	2	2
	PART B Answer Any Two Questions (Q2 x 10M = 20 M)			
1.	a) Explain Branch and Bound algorithm for Travelling salesperson problem.	5	3	2
	b)Explain the solution to the 4- queens problem using Backtracking.	5	3	2
2.	a) Write a short note on Approximation algorithms.	5	2	2
	b) Explain 3-SAT problem and prove that 3-SAT is NP Complete.	5	4	2
3.	a) Explain topological sorting with an example.	4	3	2
	b) Explain the significance of randomization in algorithms.	6	4	2